

TSIA2 Math Practice Test 6

PART A — QUESTIONS

TSIA2-M6-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student buys a lab manual for \$42 and safety goggles for \$18. Sales tax is 7% of the combined price. What is the total cost, including tax?

- A) \$60.07
- B) \$62.94
- C) \$64.20
- D) \$67.00

TSIA2-M6-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x: $2(3x + 4) - 5 = 4x + 17$

- A) 5
- B) 6
- C) 7
- D) 11

TSIA2-M6-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A trapezoid has bases of 10 inches and 16 inches and a height of 7 inches. What is the area of the trapezoid?

- A) 91 square inches
- B) 104 square inches
- C) 112 square inches
- D) 182 square inches

TSIA2-M6-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A student's test scores are 78, 84, 91, and 87. What score must the student earn on the next test to have a mean of 86 for all five tests?

- A) 86
- B) 88
- C) 90
- D) 92

TSIA2-M6-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A campus shuttle travels 7.5 miles in 18 minutes. At the same average speed, how far will it travel in 42 minutes?

- A) 14.5 miles
- B) 16.5 miles
- C) 17.5 miles
- D) 18.5 miles

TSIA2-M6-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which equation has the same solution as $5x - 9 = 21$?

- A) $5x = 12$
- B) $5x = 30$
- C) $x - 9 = 21$
- D) $x = 21 - 9$

TSIA2-M6-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular poster has an area of 300 square inches. Its width is 15 inches. What is its length?

- A) 18 inches
- B) 20 inches
- C) 25 inches
- D) 45 inches

TSIA2-M6-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A box contains 4 red markers, 5 blue markers, and 6 green markers. One marker is selected at random. What is the probability that the marker is not green?

- A) $\frac{2}{5}$
- B) $\frac{3}{5}$
- C) $\frac{5}{9}$
- D) $\frac{2}{3}$

TSIA2-M6-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

An employee earns \$17.50 per hour. Last week, the employee worked 28 hours and had \$96 withheld for taxes and deductions. What was the employee's take-home pay?

- A) \$394
- B) \$406
- C) \$490
- D) \$586

TSIA2-M6-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A table shows values of a linear function.

x: -1, 0, 1, 2

y: 7, 4, 1, -2

Which equation represents the function?

- A) $y = 3x + 4$
- B) $y = -3x + 4$
- C) $y = -4x + 3$
- D) $y = x - 3$

TSIA2-M6-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A cube has a side length of 5 centimeters. What is the volume of the cube?

- A) 25 cubic centimeters
- B) 75 cubic centimeters
- C) 100 cubic centimeters
- D) 125 cubic centimeters

TSIA2-M6-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A college advisor recorded the number of students seen each day during one week.

Monday: 14

Tuesday: 21

Wednesday: 19

Thursday: 24

Friday: 22

Which statement is true?

- A) The range is 8 students.
- B) The median is 21 students.

- C) The mean is 19 students.
- D) The mode is 22 students.

TSIA2-M6-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A savings account balance increases from \$480 to \$540 in one year. What is the percent increase?

- A) 10%
- B) 12.5%
- C) 15%
- D) 20%

TSIA2-M6-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $2x^2 + 7x - 15$?

- A) $(2x - 3)(x + 5)$
- B) $(2x + 3)(x - 5)$
- C) $(x - 3)(2x + 5)$
- D) $(x + 3)(2x - 5)$

TSIA2-M6-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A line segment has endpoints $(2, -3)$ and $(8, 5)$. What is the distance between the endpoints?

- A) 8 units
- B) 9 units
- C) 10 units
- D) 14 units

TSIA2-M6-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A student organization has 9 members from the nursing program and 6 members from the business program. If one member is chosen at random to give a presentation, what is the probability that the member is from the business program?

- A) $\frac{2}{5}$
- B) $\frac{3}{5}$
- C) $\frac{2}{3}$
- D) $\frac{3}{2}$

TSIA2-M6-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A rectangular parking permit sticker has an area of 18 square inches. If its length is 6 inches, what is its width?

- A) 2 inches
- B) 3 inches
- C) 4 inches
- D) 12 inches

TSIA2-M6-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

The value of a used tablet decreases by 12% each year. If the tablet is worth \$300 now, which expression represents its value after 2 years?

- A) $300(0.12)^2$
- B) $300(0.88)^2$
- C) $300 - 2(0.12)$
- D) $300(1.12)^2$

TSIA2-M6-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A right triangle has one leg measuring 9 feet and a hypotenuse measuring 15 feet. What is the length of the other leg?

- A) 6 feet
- B) 10 feet
- C) 12 feet
- D) 18 feet

TSIA2-M6-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A table shows the number of students who preferred two types of study sessions.

Morning session: 32 students

Evening session: 48 students

Based on the table, what percent of the students preferred the evening session?

- A) 40%
- B) 48%
- C) 60%
- D) 80%

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M6-001 — Correct: C — Correct Answer: \$64.20 — Explanation: The combined price is $\$42 + \$18 = \$60$. The sales tax is $0.07 \times 60 = \$4.20$, so the total cost is $\$64.20$.

TSIA2-M6-002 — Correct: C — Correct Answer: 7 — Explanation: Distribute and simplify: $6x + 8 - 5 = 4x + 17$, so $6x + 3 = 4x + 17$. Then $2x = 14$, so $x = 7$.

TSIA2-M6-003 — Correct: A — Correct Answer: 91 square inches — Explanation: The area of a trapezoid is $\frac{1}{2} \cdot (\text{base 1} + \text{base 2}) \cdot \text{height}$. So the area is $\frac{1}{2} \cdot (10 + 16) \cdot 7 = 91$ square inches.

TSIA2-M6-004 — Correct: C — Correct Answer: 90 — Explanation: To have a mean of 86 for 5 tests, the total must be $86 \times 5 = 430$. The first four scores total 340, so the next score must be $430 - 340 = 90$.

TSIA2-M6-005 — Correct: C — Correct Answer: 17.5 miles — Explanation: The shuttle travels $7.5 \div 18 = 5/12$ mile per minute. In 42 minutes, it travels $42 \times 5/12 = 17.5$ miles.

TSIA2-M6-006 — Correct: B — Correct Answer: $5x = 30$ — Explanation: Adding 9 to both sides of $5x - 9 = 21$ gives the equivalent equation $5x = 30$.

TSIA2-M6-007 — Correct: B — Correct Answer: 20 inches — Explanation: For a rectangle, area = length $\times$ width. Since $300 = \text{length} \times 15$, the length is $300 \div 15 = 20$ inches.

TSIA2-M6-008 — Correct: B — Correct Answer: $3/5$ — Explanation: There are 15 markers total, and 9 are not green. The probability is $9/15 = 3/5$.

TSIA2-M6-009 — Correct: A — Correct Answer: \$394 — Explanation: Gross pay is $17.50 \times 28 = \$490$. After \$96 is withheld, take-home pay is $\$490 - \$96 = \$394$.

TSIA2-M6-010 — Correct: B — Correct Answer: $y = -3x + 4$ — Explanation: The y-values decrease by 3 for each increase of 1 in x, so the slope is -3 . When $x = 0$, $y = 4$, so the equation is $y = -3x + 4$.

TSIA2-M6-011 — Correct: D — Correct Answer: 125 cubic centimeters — Explanation: The volume of a cube is side^3 . With side length 5, the volume is $5^3 = 125$ cubic centimeters.

TSIA2-M6-012 — Correct: B — Correct Answer: The median is 21 students. — Explanation: In order, the values are 14, 19, 21, 22, and 24. The middle value is 21, so the median is 21.

TSIA2-M6-013 — Correct: B — Correct Answer: 12.5% — Explanation: The increase is $\$540 - \$480 = \$60$. The percent increase is $60/480 = 0.125$, or 12.5%.

TSIA2-M6-014 — Correct: A — Correct Answer: $(2x - 3)(x + 5)$ — Explanation: Multiplying gives $2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.

TSIA2-M6-015 — Correct: C — Correct Answer: 10 units — Explanation: The horizontal change is $8 - 2 = 6$, and the vertical change is $5 - (-3) = 8$. The distance is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ units.

TSIA2-M6-016 — Correct: A — Correct Answer: $2/5$ — Explanation: There are $9 + 6 = 15$ members total, and 6 are from the business program. The probability is $6/15 = 2/5$.

TSIA2-M6-017 — Correct: B — Correct Answer: 3 inches — Explanation: For a rectangle, area = length $\times$ width. Since $18 = 6 \times \text{width}$, the width is $18 \div 6 = 3$ inches.

TSIA2-M6-018 — Correct: B — Correct Answer: $300(0.88)^2$ — Explanation: A 12% yearly decrease means the value is multiplied by $1 - 0.12 = 0.88$ each year. After 2 years, the value is $300(0.88)^2$.

TSIA2-M6-019 — Correct: C — Correct Answer: 12 feet — Explanation: Using the Pythagorean theorem, the missing leg is $\sqrt{(15^2 - 9^2)} = \sqrt{(225 - 81)} = \sqrt{144} = 12$ feet.

TSIA2-M6-020 — Correct: C — Correct Answer: 60% — Explanation: There are $32 + 48 = 80$ students total. The evening session was preferred by 48 students, and $48/80 = 0.60$, or 60%.